FairChild Institute

Data Science Institute

12-Month Practical Data Science Mentorship Program

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Month 1

Data Science Thinking & Python Foundations

Phase: FOUNDATION | Theme: From curious learner to computational problem solver

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Data science workflow and problem framing	What is the decision, and what would count as useful evidence?	Build and verify a small reproducible artifact demonstrating data science workflow and problem framing.
2	Python types, expressions and notebooks	How can I represent a problem in code?	Build and verify a small reproducible artifact demonstrating python types, expressions and notebooks.
3	Control flow, functions and debugging	How can I debug a solution systematically?	Build and verify a small reproducible artifact demonstrating control flow, functions and debugging.
4	Collections, files, Git and Python mini-project	How do I make a small analysis reproducible?	Build and verify a small reproducible artifact demonstrating collections, files, git and python mini-project.

Final Project	Python notebook analysing a small public dataset

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Month 2

Mathematics, Probability & Data Reasoning

Phase: FOUNDATION | Theme: From code execution to quantitative reasoning

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Data science workflow and problem framing	What does the geometry of data tell us?	Build and verify a small reproducible artifact demonstrating vectors, matrices and numerical reasoning.
2	Functions, derivatives and gradients conceptually	Why do models optimise functions?	Build and verify a small reproducible artifact demonstrating functions, derivatives and gradients conceptually.
3	Probability, randomness and simulation	How does randomness affect conclusions?	Build and verify a small reproducible artifact demonstrating probability, randomness and simulation.
4	Descriptive statistics and distributions	How do I summarise variation honestly?	Build and verify a small reproducible artifact demonstrating descriptive statistics and distributions.

Final Project	Simulation and exploratory numerical reasoning notebook

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Month 3

SQL & Relational Data Analysis

Phase: DATA FOUNDATIONS | Theme: From flat files to relational evidence

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Relational data, schemas and keys	How is business information organised across tables?	Build and verify a small reproducible artifact demonstrating relational data, schemas and keys.
2	SQL filtering, sorting and aggregation	How do I translate questions into queries?	Build and verify a small reproducible artifact demonstrating sql filtering, sorting and aggregation.
3	Joins, CTEs and analytical windows	How do I combine tables without corrupting counts?	Build and verify a small reproducible artifact demonstrating joins, CTEs and analytical windows.
4	SQL business case and reconciliation	How can I prove SQL results are correct?	Build and verify a small reproducible artifact demonstrating sql business case and reconciliation.

Final Project	SQL case study with reconciled multi-table metrics

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Month 4

Data Preparation, Pipelines & EDA

Phase: DATA FOUNDATIONS | Theme: From raw records to trustworthy evidence

Duration 4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	pandas cleaning and data quality	What makes raw data unsuitable for analysis?	Build and verify a small reproducible artifact demonstrating pandas cleaning and data quality.
2	Combining datasets and feature construction	How can preparation be repeated reliably?	Build and verify a small reproducible artifact demonstrating combining datasets and feature construction.
3	Exploratory analysis and visualisation	Which patterns are meaningful and which misleading?	Build and verify a small reproducible artifact demonstrating exploratory analysis and visualisation.
4	EDA report and data pipeline mini-project	How do I document and test a data pipeline?	Build and verify a small reproducible artifact demonstrating EDA report and data pipeline mini-project.

Final Project Reproducible cleaning pipeline and exploratory report

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Month 5

Statistics, Inference & Uncertainty

Phase: STATISTICAL REASONING | Theme: From descriptive claims to defensible inference

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Sampling and statistical uncertainty	What can a sample tell us about a population?	Build and verify a small reproducible artifact demonstrating sampling and statistical uncertainty.
2	Confidence intervals and estimation	How precise is an estimate?	Build and verify a small reproducible artifact demonstrating confidence intervals and estimation.
3	Hypothesis testing and effect size	When is an observed difference convincing?	Build and verify a small reproducible artifact demonstrating hypothesis testing and effect size.
4	Assumptions, pitfalls and statistical report	Which claims remain uncertain?	Build and verify a small reproducible artifact demonstrating assumptions, pitfalls and statistical report.

Final Project	Statistical investigation with confidence intervals and assumption checks

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Month 6

Regression, Causality & Experiments

Phase: STATISTICAL REASONING | Theme: From association to careful explanation

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Linear regression and diagnostics	What does a regression model actually explain?	Build and verify a small reproducible artifact demonstrating linear regression and diagnostics.
2	Logistic regression and interpretation	How should classification effects be interpreted?	Build and verify a small reproducible artifact demonstrating logistic regression and interpretation.
3	Causality, confounding and limitations	Why is correlation not necessarily causation?	Build and verify a small reproducible artifact demonstrating causality, confounding and limitations.
4	A/B test design and experimental analysis	How should an experiment be designed?	Build and verify a small reproducible artifact demonstrating a/b test design and experimental analysis.

Final Project	Regression study and pre-registered A/B test design

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Month 7

Supervised Machine Learning Foundations

Phase: PREDICTIVE MODELLING | Theme: From fitting algorithms to building baselines

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	ML framing, targets and baseline models	What exactly should a model predict?	Build and verify a small reproducible artifact demonstrating ml framing, targets and baseline models.
2	Splits, cross-validation and leakage	How do I prevent leakage and optimistic scores?	Build and verify a small reproducible artifact demonstrating splits, cross-validation and leakage.
3	Linear models and regularisation	How do simple models become useful baselines?	Build and verify a small reproducible artifact demonstrating linear models and regularisation.
4	Tree models and comparative evaluation	Which model performs better and for whom?	Build and verify a small reproducible artifact demonstrating tree models and comparative evaluation.

Final Project	Baseline supervised prediction and leakage-safe evaluation

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Month 8

Model Evaluation, Tuning & Unsupervised ML

Phase: PREDICTIVE MODELLING | Theme: From model score to evidence of generalisation

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Ensembles and feature importance	How do I interpret a complex model responsibly?	Build and verify a small reproducible artifact demonstrating ensembles and feature importance.
2	Classification metrics and threshold choice	Which metric fits the decision?	Build and verify a small reproducible artifact demonstrating classification metrics and threshold choice.
3	Preprocessing pipelines and tuning	Can tuning be trusted on unseen data?	Build and verify a small reproducible artifact demonstrating preprocessing pipelines and tuning.
4	Unsupervised learning and ML course project	When does unsupervised learning help?	Build and verify a small reproducible artifact demonstrating unsupervised learning and ml course project.

Final Project	Compared and tuned ML pipeline with error analysis

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Month 9

Forecasting, Robustness & Applied ML

Phase: ADVANCED APPLICATION | Theme: From one-off model to realistic validation

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Error analysis and model calibration	Where is the model failing?	Build and verify a small reproducible artifact demonstrating error analysis and model calibration.
2	Class imbalance and robustness	Does performance survive real-world variation?	Build and verify a small reproducible artifact demonstrating class imbalance and robustness.
3	Time series and temporal validation	How does time change validation?	Build and verify a small reproducible artifact demonstrating time series and temporal validation.
4	Forecasting applied project	Is the forecast better than a simple baseline?	Build and verify a small reproducible artifact demonstrating forecasting applied project.

Final Project	Time-aware forecasting project and robustness report

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Month 10

Power BI Delivery & Cross-Tool QA

NLP, Deep Learning & Responsible AI

Phase: ADVANCED APPLICATION | Theme: From novel methods to responsible model choices

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Text analytics and NLP basics	How can machines represent text?	Build and verify a small reproducible artifact demonstrating text analytics and nlp basics.
2	Recommender systems overview	What are recommendation trade-offs?	Build and verify a small reproducible artifact demonstrating recommender systems overview.
3	Deep learning fundamentals and boundaries	When is deep learning justified?	Build and verify a small reproducible artifact demonstrating deep learning fundamentals and boundaries.`
4	Fairness, privacy and responsible ML review	Who might be harmed by the model?	Build and verify a small reproducible artifact demonstrating fairness, privacy and responsible ml review.

Final Project	Time-aware forecasting project and robustness report

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Month 11

MLOps, Deployment & Professional Practice

Phase: DELIVERY | Theme: From notebook to reproducible handover

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Text analytics and NLP basics	How can another analyst reproduce this result?	Build and verify a small reproducible artifact demonstrating experiment tracking and reproducibility.
2	Recommender systems overview	How can a model be used safely?	Build and verify a small reproducible artifact demonstrating packaging and serving models.
3	Deep learning fundamentals and boundaries	How would we detect degradation?	Build and verify a small reproducible artifact demonstrating monitoring and drift awareness.
4	Fairness, privacy and responsible ML review	Can I hand this work to another team?	Build and verify a small reproducible artifact demonstrating documentation, model cards and portfolio review.

Final Project	Packaged inference demo, monitoring plan and portfolio case

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Month 12

Capstone: End-to-End Data Science Solution

Independent Data Science Capstone

Phase: CAPSTONE | Theme: From guided learner to independent data scientist

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Capstone discovery and data audit	What is the real decision and ethical scope?	Build and verify a small reproducible artifact demonstrating capstone discovery and data audit.
2	Baseline, modelling and validation	What is the simplest defensible baseline?	Build and verify a small reproducible artifact demonstrating baseline, modelling and validation.
3	Prototype, documentation and stakeholder review	Can a stakeholder use and trust the prototype?	Build and verify a small reproducible artifact demonstrating prototype, documentation and stakeholder review.
4	Capstone presentation, viva and portfolio packaging	Can I defend every important choice?	Build and verify a small reproducible artifact demonstrating capstone presentation, viva and portfolio packaging.

Final Project	End-to-end data science capstone and oral defence